Project Brief - Analyzing the Indonesian Supermarket Business

Project Owner: Muchammad Wildan Alkautsar

Executive Summary

This project analyzes Indonesian supermarket transaction data from 2014 to 2017 to address limited visibility into sales performance, regional demand, product contributions, shipping preferences, and profit drivers. The dataset will be cleaned and analyzed using Google Sheets, visualized through an interactive Looker Studio dashboard, and used to develop regression models in Orange Data Mining. The expected outcomes are reliable business insights, an evaluated profit-prediction model, and actionable recommendations to support the company's sales, inventory, marketing, and operational decisions.

Background & Objective

A leading Indonesian supermarket has transaction and customer data from 2014 to 2017, including sales, profit, products, shipping, order dates, and customer locations. However, the dataset contains duplicate records, missing values, and inconsistent city formats that must be cleaned before analysis.

The company lacks reliable insights into sales trends, regional order distribution, product category contributions, top-performing cities, preferred shipping methods, and peak sales months. Data quality issues must first be resolved to ensure accurate analysis.

The project will produce a cleaned dataset, Google Sheets query results, an interactive Looker Studio dashboard, and a presentation summarizing key findings. These outputs will help the CEO make informed decisions about sales strategy, product priorities, regional opportunities, and shipping operations.

Analytical Questions

1. What was the trend in sales value from 2014 to 2017?
2. What was the proportion of orders by region?
3. What was the sales proportion for each product category?
4. Which city generated the highest sales?
5. Which shipping method was most frequently used by buyers?
6. Which month saw the highest sales between 2014 and 2017?

Personnel Involved

Stakeholder List

- **Chief Executive Officer (CEO) - Approver**; approves the project objectives and uses the final insights for strategic decision-making.
- **Sales Manager** - **Informee**; reviews sales trends, regional performance, and product category results.
- **Operations Manager** - **Informee**; reviews shipping method preferences and operational findings.

Contributor List

- **Data Analyst** - **Project Lead**; cleans and validates the dataset, performs analysis using Google Sheets `QUERY()`, builds the Looker Studio dashboard, and presents recommendations.
- **Sales Data Administrator** - **Data Provider**; provides transaction data and clarifies sales, product, customer, and shipping fields.
- **Business Intelligence Analyst** - **Dashboard Reviewer**; reviews dashboard accuracy, visualization clarity, and KPI consistency.

Expected Deliverable

The project will produce two primary deliverables:

- **Predictive Modeling with Orange Data Mining:** A saved Orange workflow named `analisis-prediktif-supermarket.ows` that uses cleaned retail data to predict profit. The workflow includes feature selection, three regression models, model evaluation using RMSE and R^2, and a comparison of actual and predicted values.
- **Interactive Dashboard with Google Looker Studio:** A dashboard that presents sales trends from 2014 to 2017, orders by region, sales by product category, the highest-performing city and month, and the most frequently used shipping method.

Timeline & Milestone

The project will be completed within one day through the following milestones:

- **Milestone 1: Data Preparation** - **08:00–10:00**; remove duplicates, handle missing values, standardize city names, and validate the cleaned dataset.
- **Milestone 2: Data Analysis and Dashboard** - **10:00–13:00**; answer analytical questions using Google Sheets `QUERY()` and build the Looker Studio dashboard.
- **Milestone 3: Predictive Modeling** -**13:00–16:00**; create and evaluate three regression models in Orange Data Mining using RMSE and R^2.
- **Milestone 4: Final Review and Submission** -**16:00–18:00**; verify all results, finalize the presentation, and organize the dashboard, workflow, and supporting links.

Communication Plan

Stakeholders will receive concise progress updates throughout the one-day project using meetings, email, and shared project links.

- **Project Kickoff Meeting** -Confirm project objectives, analytical questions, and expected deliverables; once at the beginning of the project.
- **Progress Check-in** - Report data-cleaning, dashboard, and modeling progress; twice during the project.
- **Email or Chat Updates** -Communicate completed milestones, issues, and required decisions; as needed.
- **Shared Project Tracker** -Record task status, deliverable links, and completion progress; updated after each milestone.
- **Final Review Meeting** -Present dashboard insights, model performance, conclusions, and recommendations; once at the end of the project.

Further Resource & Reference

- **Looker Studio Dashboard**: https://datastudio.google.com/reporting/bd8c4a34-bf0b-47e3-a728-d2ec0e7424bc
- **Dataset:** https://docs.google.com/spreadsheets/d/1kAcBhvVvTxYVrEivmiBagRVlj1USuxnqEsJIYNNQzCw/copy